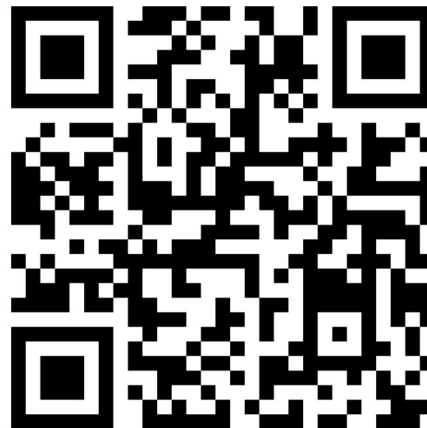


3 exercises for HealthcareNLP tutorial

open source websites



<https://github.com/4dpicture/HealthNLP>



<https://arxiv.org/abs/2512.08617>



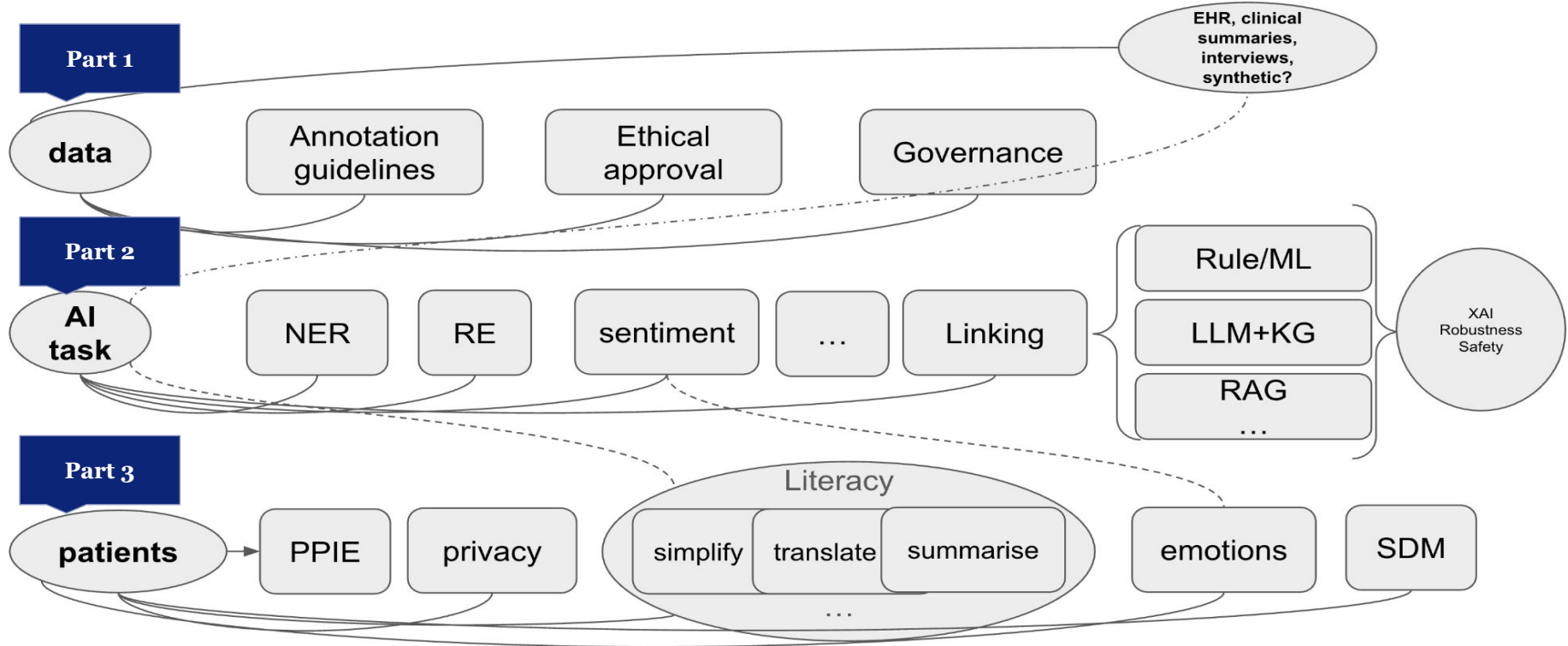
Morning session:
9:00 to 13:00
Morning Coffee
break: 10.30-11.00

LREC 2026

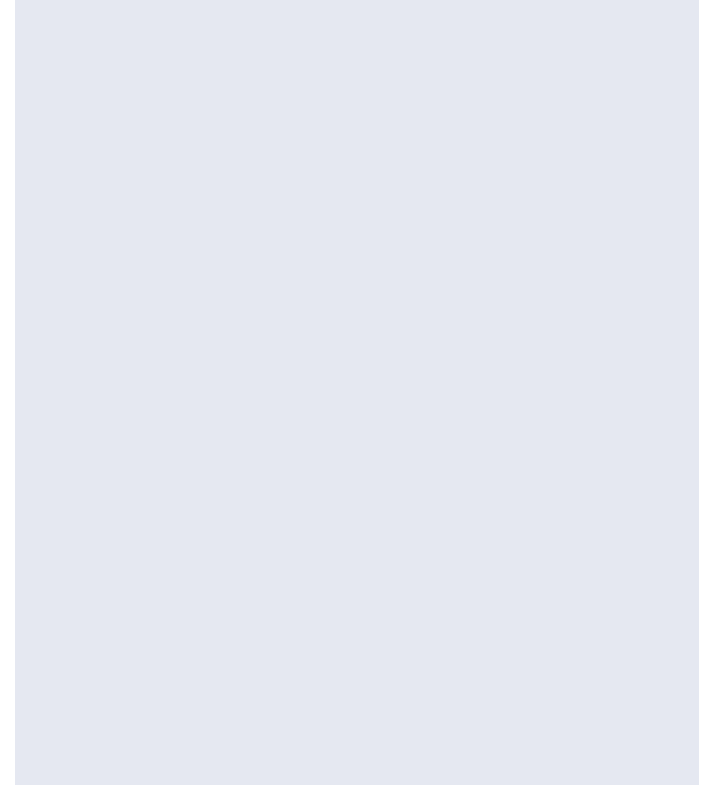
Schedule Tuesday, 12 May 2026

	Morning session	Afternoon session
Room 1	<u>W2</u> The Third Workshop on Patient-oriented Language Processing (CL4Health)	
Room 2	<u>W36</u> SIGUL 2026 Joint Workshop with ELE, EURALI, and DCLRL: Towards Inclusivity and Equality: Language Resources and Technologies for Under-Resourced and Endangered Languages	<u>W31</u> The Seventh Workshop on Resources for African Indigenous Languages (RAIL)
Room 3	<u>W44</u> Joint Workshop on Legal and Ethical Issues in Human Language Technologies (LEGAL2026) & Computational Approaches to Language Data Pseudonymization, Anonymization, De-identification, and Data Privacy (CALD-pseudo 2026)	
Room 4	<u>W26</u> NLPerspectives: The 5th Workshop on Perspectivist Approaches to Natural Language Processing	
Room 5	<u>T15</u> Hallucination in Large Foundation Models	<u>W45</u> The Information Disorder Workshop
Room 6	<u>W15</u> 3rd International Workshop on Natural Scientific Language Processing	
Room 7	<u>T13</u> HealthcareNLP: where we are and what's next	<u>W13</u> Resources and Processing of linguistic, para-linguistic and extralinguistic Data from people with various forms of cognitive/psychiatric/ developmental impairments (RaPID-6@MENTAL.ai@LREC)

structure of the tutorial



Exercise I



Example task: what do we need

Task: find side effects for medications in online cancer patient forum discussions

How would you approach this? What do you need?

Discuss with your neighbour for a few minutes



yes, bad chills at the beginning of my treatment that I end up at emergency, this happened about three times . After two years taken Gleevec still very cold, but not too bad.

The water retention is a pain in the ass . Can you do sports? It's what helps me most.

Example task: what do we need

Steps to take

1. Filter the potentially relevant messages
2. Get/create training data for NER
3. Train an NER model to identify drug names and side effects in the messages
4. Normalize the side effects (map to ontology)
5. Relation extraction: cooccurrences of drug names and side effects in one message
6. (Match the found relations to an existing knowledge base to identify which relations are new)

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- Lists/ontologies of drug names and known side effects (e.g. SNOMED CT)
- Pre-processing tools
- Pre-trained BERT models for NER and ontology linking
- Labelled data for supervised NER finetuning and evaluation

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Exercise: annotation for NER

Task: mark in the following messages all strings (single- or multi-words) that refer to:

- A **disease**
- A **medication**
- A **side effect**

Use 3 different colors for your markup

You can find the PDF on our git repository

After your annotation, discuss with your neighbour



#	post
1	I was only on Gleevec for 1 year so I guess about 3 years later.
2	I'm not sure what you mean by severe but while taking gleevec I was prone to eye bleeds, more than just blood shot but bright red . It was only in my left eye and at the same time I was being seen by the eye hospital about something else . They new about my medication and gave my eyes a good examination but could find no reason for the bleeding and I had no cause to worry . It has stopped since I stopped taking gleevec . I'm not saying this is the same in your case so get it checked out just in case.
3	Most important: Consult an oncologist who is knowledgeable and experienced with GIST, specifically . Ask lots of questions, and keep asking until you get the answers you understand.
4	yes, bad chills at the beginning of my treatment that I end up at emergency, this happened about three times . After two years taken Gleevec still very cold, but not too bad.
5	Someone asked me yesterday how cancer has changed my life . You said it much better than I did! Thanks.
6	Those cysts are quite common in just about everyone . However us GIST warriors it makes us worry . Always good to check and ask for an opinion.
7	The water retention is a pain in the ass . Can you do sports? It's what helps me most.
8	Votrient is another cancer drug that has been used off label for GIST with mixed results . I was on Stivarga also, it made my BP skyrocket . Fornunately, my primary care physican is a good friend and has taken over the managing of BP meds for me . It is great to have a doctor who gives you their personal cell phone number and will take your call or text even on the weekend! I am blessed.

Exercise II

Exercise: entity extraction

In **settings without in-domain labelled data** data we have two options:

- use of a BERT checkpoint that someone else fine-tuned for a similar task
- prompting a generative LLM to do the task

In this exercise you will compare both on a medical entity recognition task, using the same small sample of forum messages that you manually annotated before.

1. For running the BERT model, we have prepared this colab notebook:

https://colab.research.google.com/drive/14XJouUPjHE9scJcK9J4pz_qEf-GwbMj0?usp=sharing

2. For running the generative LLM, you can use this privacy-preserving interface, where you can select your own favourite model: <https://duck.ai/chat?ia=chat&duckai=1> (prompts on the next slide)

Exercise: entity extraction

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Prompt suggestion:

Please extract medical entities from the following texts. In the output, first repeat the sentence, then list the extracted entity spans (one per line), and label them with one of the following categories: MEDICATION BIOLOGICAL_STRUCTURE SIGN_SYMPTOM CLINICAL_EVENT SEVERITY DATE DURATION DISEASE_DISORDER DIAGNOSTIC_PROCEDURE
+ All 8 texts

Exercise: entity extraction

3. Compare both the BERT model output and the LLM output to your own annotations.

Which modelling approach is better?

What are problems for both models?

4. Try out another model from the same interface (e.g. replace GPT-5 by GPT-OSS or Mistral)

5. Do multiple runs with the same model to investigate consistency.

Discuss with your neighbour

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Votrient is another cancer drug that has been used off label for GIST with mixed results .

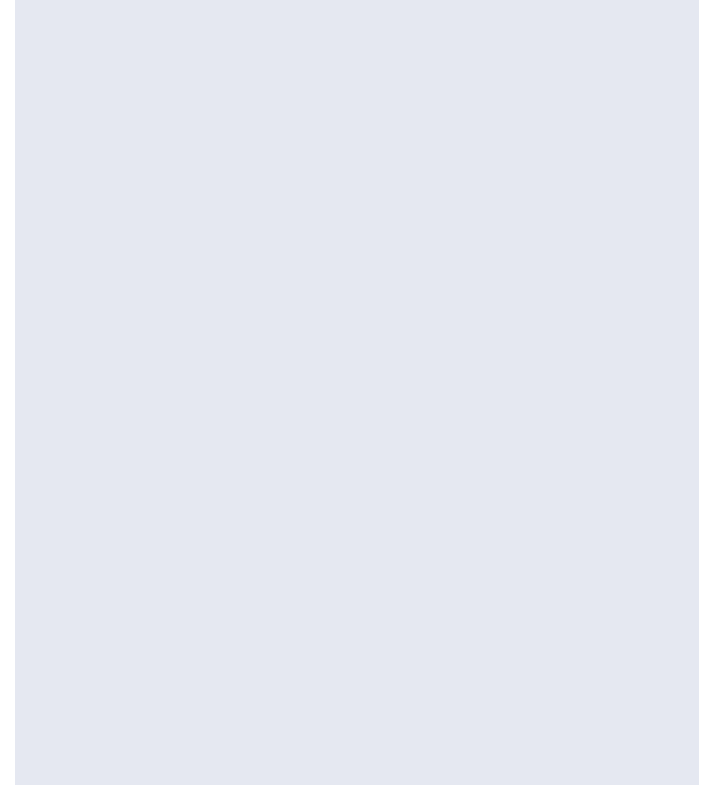


 GPT-5 mini

Votrient is another cancer drug that has been used off label for GIST with mixed results .

Votrient — MEDICATION
cancer — DISEASE_DISORDER
off label — CLINICAL_EVENT
GIST — DISEASE_DISORDER
mixed results — SIGN_SYMPTOM

Exercise III: discussion and more on hand-on



Discussion-1: AI for Reflexive QR

Jowsey, T., Braun, V., Clarke, V., Lupton, D., & Fine, M. (2025). We reject the use of generative artificial intelligence for reflexive qualitative research. *Qualitative Inquiry*, 10778004251401851.
(Position/opinion paper: 1.5 pages)

(participants download this paper pre-Tutorial?)

=> how do you think of this paper's point?

=> do/will you do "reflexive QR", if or not to use AI?

Discussion-1: AI for Reflexive QR

1) What is Reflexive Qualitative Research?

Reflexive Qualitative Research (RQR) is an approach where:

Knowledge is *co-constructed* by the researcher and the data through interpretation and reflection.

It is **not** about extracting “objective facts” from text.
Instead, it asks:

- What meanings do participants construct?
- How do I (as researcher) interpret those meanings?
- How does my background influence what I see?

So the researcher is **part of the analytic instrument**, not a neutral observer.

Discussion-1: AI for Reflexive QR

The paper outlines three primary objections:

1. **GenAI Cannot Make Meaning**

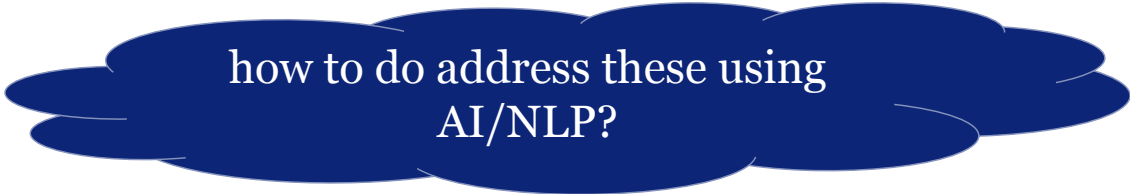
- The authors argue that simulated intelligence *does not and cannot engage in authentic meaning-making*, a central task in qualitative analysis.

2. **Qualitative Analysis Should Be a Human Practice**

- They assert that reflexive qualitative work is inherently human — it involves interpretation, positioning, and ethical engagement with data — attributes they believe AI cannot possess or perform.

3. **Ethical & Justice Concerns**

- The paper cites *broader ethical issues*: environmental impact of GenAI technologies, labor and extraction concerns (especially relating to Global South contributors), and methodological justice.



how to do address these using
AI/NLP?

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Hint: Agent-based learning,
Persona, Meaning, interpretable,
Smarter-instead-of-larger?

Hands-on Options

INSIGHTBUDDY-AI: Medication Extraction and Entity Linking using Large Language Models and Ensemble Learning

<https://github.com/HECTA-UoM/InsightBuddy-AI>

Learning Entities from Narratives of Skin Cancer (LENS)

<https://github.com/4dpicture/LENS>

MedCAT Tutorials

<https://github.com/CogStack/MedCATtutorials>